Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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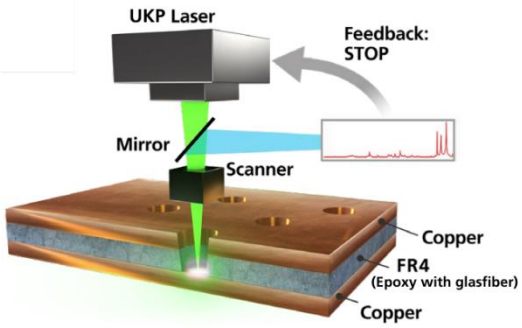
2. Technology Overview

Title of Proposed Technology	Development of a direct feedback system for laser micro machining and laser via drilling based on plasma spectroscopy			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), ~~Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )~~			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	Laser Processing Machines	Semiconductor Equipment	Measurement/Inspection Equipment
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology*			

Laser micro machining is highly used in electronics and semiconductor industries to structure conductor tracks or to drill conducting vias. Quality control is essential, as the starting materials of circuit boards have high tolerance. As production speed is very high this quality control can only be done after production, sometimes at the fully assembled PCB either with functional testing or with microscope images of random samples from the production.



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	We suggest monitoring the created plasma at the laser micro machining in real time. This is possible due to the selection of single, very narrow optical emission lines (0.1 nm spectral width), highly sensitive, amplified, time resolved detectors (<10 ns time resolution) and analog signal processing. The signal derived from each ablation pulse can be used to determine the process parameters for the next laser pulse. The signal processing is done in real time e.g. monitoring 2 MHz laser repetition rates. This enables 100% quality control, direct feedback on the production process, short setup time for new materials and quality documentation. Integrated in machining for laser drilling of microvias, this enables the production of multilayer PCBs with 100% quality control inline and reduces following test procedures.
Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* Fraunhofer IPM develops optical measurement techniques for production control like plasma spectroscopy. IPM needs a partner in the electronics industry to implement the technology in its application. This partner needs expertise in testing equipment, production processes of electronics and laser micro machining systems. The joined project partners can develop a prototype fully functional for electronics industries.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 1. Optical process control for laser micro machining 2. Inline quality control 3. Spectroscopy of laser-induced plasma 4. Pulse resolved direct feedback

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5)	End of the Project	Level (7)
R&D Objectives	Describe the specific technologies you aim to develop through this R&D project Laser ablation processes always create very weak plasma emissions and thermal radiation. The plasma emission has a typical lifetime of 100ns (for UKP laser processes), is so weak (per single pulse) that it often cannot be detected with spectrometers and is superimposed by thermal radiation. The aim is to develop a fast, sensitive sensor to monitor the plasma decay of the atomic emission in real time and derive a control signal for laser control.			
R&D Target Specifications	Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters • control signal showing the copper (or other metal) content in the laser plasma at up to 1 MHz laser repetition rate • Fast electronics with analog (quantity) and digital (copper yes/no) feedback signal at 1 MHz • detection for typical laser pulse energy: 500 fs, 20µJ, 10µm focal size			
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* The suggested project includes the following work packages: **WP0** (M1-M6): Pre-study: Specification of micro machining parameters and conditions. Definition of processed materials. Tests with real samples and reproduced laser conditions at IPM to prove the feasibility of the technology. **WP1** (M7-M18): Development of an optical sensor for quality control in laser micro machining processes (e.g. Via drilling in electronics, laser micromachining for FIB preparation, laser removal on multi-layer materials): • Design of an optical system collecting the created Plasma light of an ultra-short pulsed laser process (e.g. 1 – 2000 kHz, 1fs-10ns laser) and detecting single elements, which show material change (e.g. Copper – FR4 at PCB in electronics). **WP2** (M9-M18): Fast electronics: • Develop a fast electronic circuit to evaluate the plasma signal in realtime and create a direct feedback signal to control the laser process. • Signal storage with FIPO for quality documentation **WP3** (M10-M18): Interface • optical interface before galvo-scanner to guide the plasma light to the detection fiber/sensor • electrical interface: Laser trigger, feedback to processing software, data storage for quality assessment **Milestone:** Sensor and interface are ready for integration. Sensor works at speed of 1MHz and detects plasma signal at 10µJ pulse energy. **WP4** (M19-M30): Integration • Integration of the sensor into an existing laser micro machining process • Testing of direct feedback loop to the laser to stop the ablation process when quality criteria are met (at up to 2 MHz laser repetition rate)			

	=> technology to enable inline control and regulation for laser ablation processes in production of (micro-) electrical circuits **WP5** (M31-M36): Adoption and Customer demands: • Adjustment according to lessons learned from the integration phase • Discussion with customers according to their application and needs • Possible adoption to customers

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* Fraunhofer IPM develops the sensor to be integrated in the micromachining system: • Optical design and hardware contribution for laser micromachining system • Participation in the integration into a working prototype. • Participation in the development and refinement of the feedback loop and interpretation of the data to ensure rapid transfer from lab to market-level prototype. • Cooperation in transfer to series production, cooperation in further development for customized systems
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* • Creation of an optical interface in the laser micromachining system • Definition of the laser ablation process and the processed materials • Development of a feedback loop accepting and processing the data from the sensor • Testing and piloting at a production facility

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* • Quality control system integrated into a laser micro machining system • System design ready for production at high quantities • Drilling process is faster as no surcharge is needed and always ablates the right amount of material • Set-up process requires no time-consuming steps with laboratory work under the microscope • System enabling fast development of new production processes for electronic devices (usable for R&D groups in production industry) • No production waste due to 100% quality control and cost saving due to early stage monitoring

6. References

(1)

(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
begin	**Prototype Stage (5-6)**	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
end	**Productization Stage (7-8)**	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization